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Agenda

Read Satellite Images From The earthaccess Library

Optimize Prithvi to Accurately Read Corn Data

Predict Corn Yield Based on ML Prediction

Export Information to a Descriptive GUI

Why Prompt 2?

- Machine learning familiarity
- Sustainability/reusability
  - Prithvi is a great pretrained model, just not utilized for corn data
  - Saves resources
- Interested in the prompt

State Boundaries

- To acquire the correct state data, geojson files set the boundaries from the satillite images

- {"type":"FeatureCollection","features":[{"type":"Feature","properties":{},"geometry":{"coordinates":[[[-102.03307524801697, 41.0117612332937],[-102.03340659521527, 37.017916855477324],[-109.04188443282699, 37.018112527549576],[-109.04174851415938, 41.02508928997431],[-102.03307524801697, 41.0117612332937]]],"type":"Polygon"}}]}
  - The boundaries for Colorado

- These geojson files allow us to ensure that images only contain data from the given states.

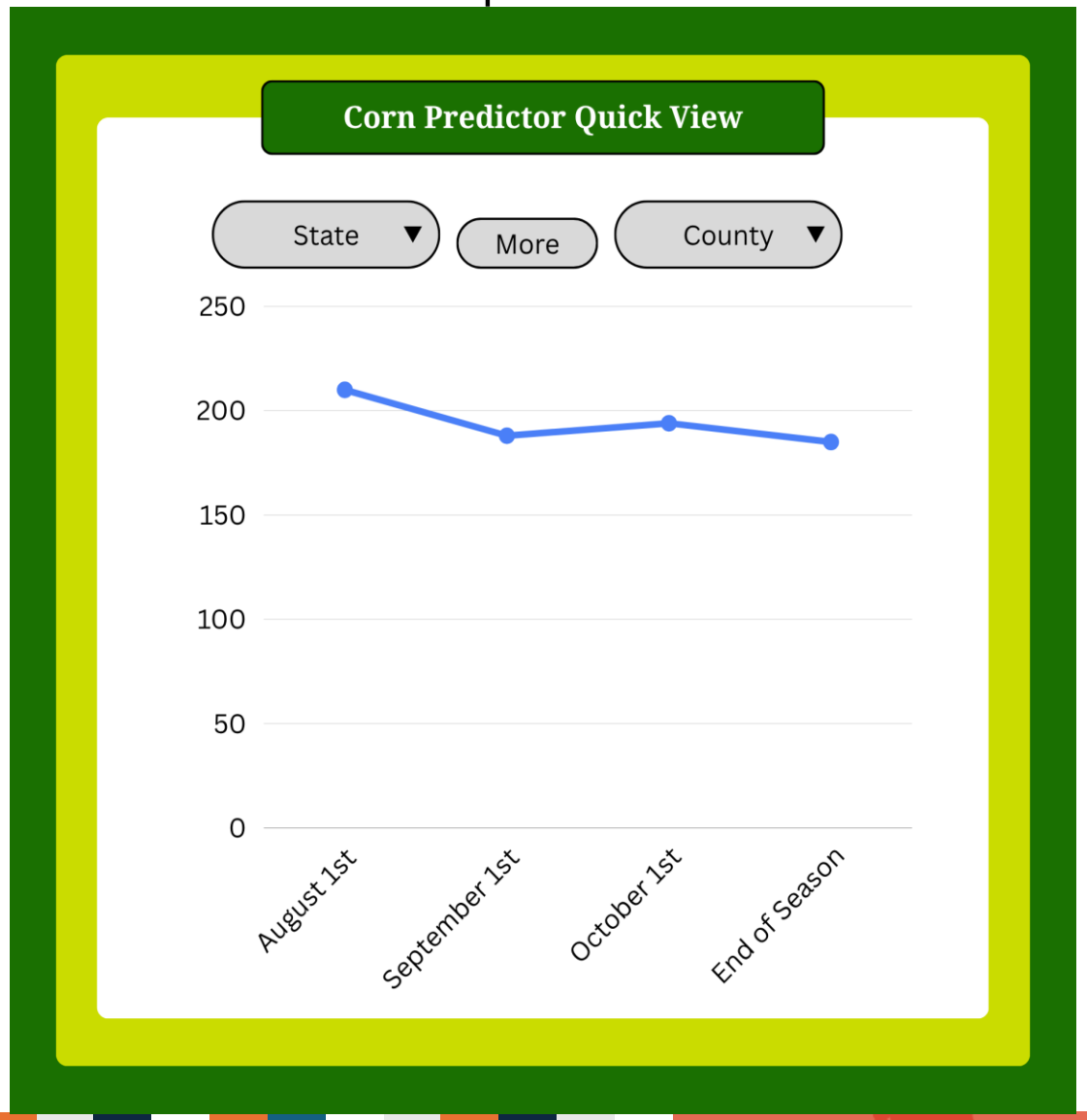
About Our Model

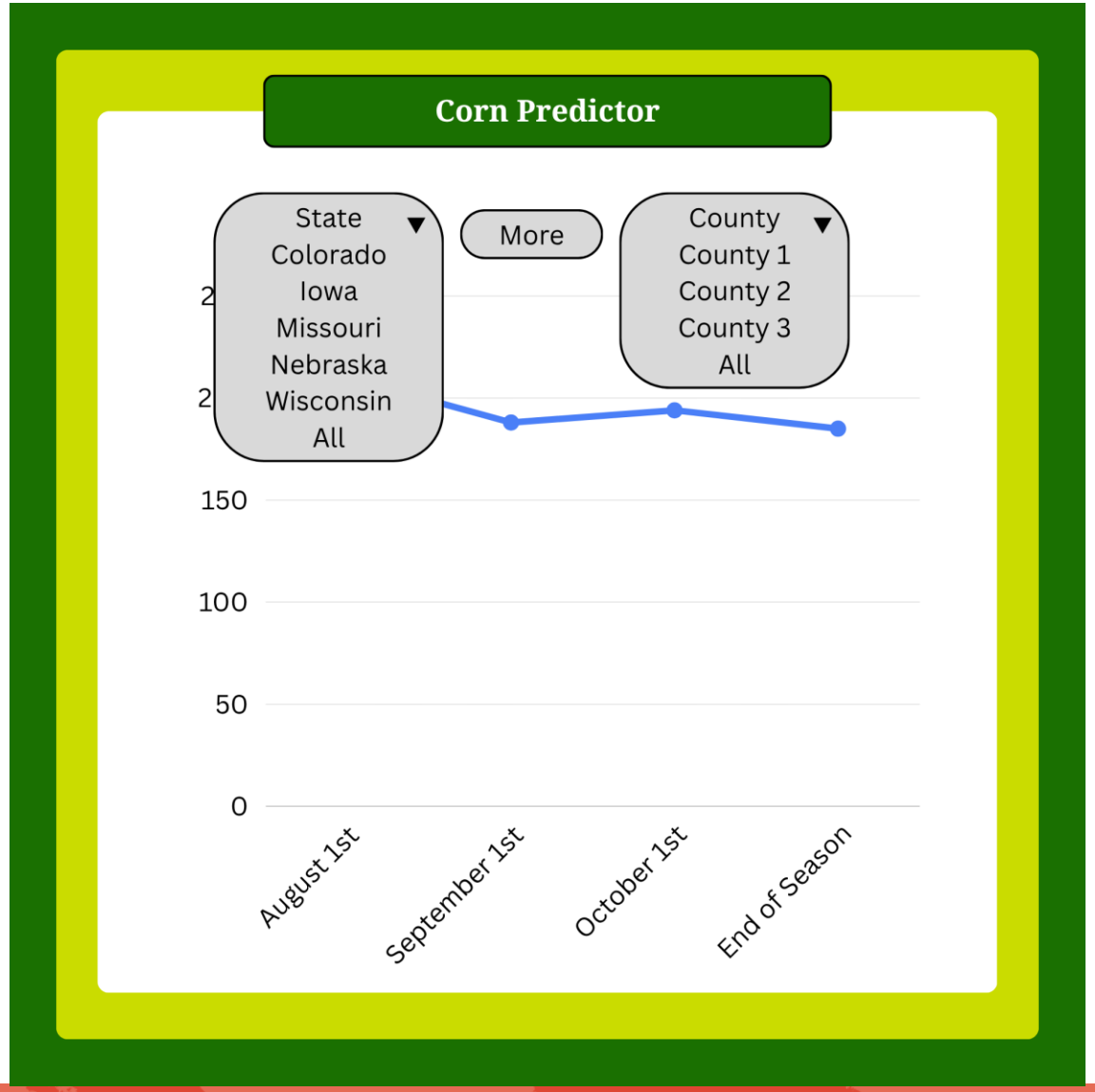
- Input is a csv containing desired states and their data
- Reads satellite images from earthaccess
- Takes metadata from the files to the model that performs predictions

If given more time we would:

- Fully expand the model to predict outside of Colorado
- Finish a GUI (on next slide)

Intended Final Output





Intended Final Output Continued

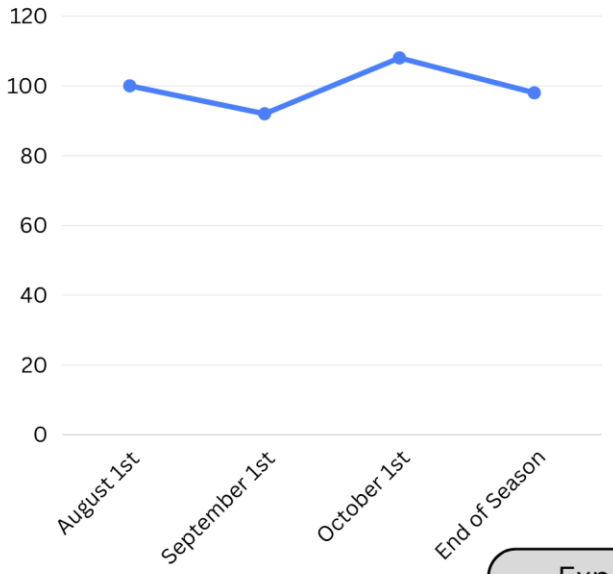
Douglass County, CO Detailed Report

August 1st Yield: 100 Bushels per Acre, +-5.0%

September 1st Yield: 92 Bushels per Acre, +-3.4%

October 1st Yield: 108 Bushels per Acre +-2.5%

October 20th Yield: 98 Bushels per Acre +-2.0%
(End of Season)



Export To CSV

Project Challenges

- Initial team issues
- Importing data
- Prithvi implementation
- Prediction model